Following The Prompt: An empirical analysis of AI traffic routing across heterogeneous ISPs

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1 Research Question

A handful of AI services (e.g. ChatGPT, Gemini, and Claude) now mediate a rapidly growing share of everyday queries, yet the network paths that carry those queries are almost entirely invisible to users. We ask three questions whose answers have direct implications for data sovereignty, surveillance exposure, and resilience:

- **RQ1 (Path Divergence):** How does the choice of local American ISPs (e.g., Comcast, T-Mobile, Verizon, etc.) influence the path taken by traffic destined for major AI endpoints?
- **RQ2 (Infrastructural Convergence):** Despite divergent domestic starting points and transport mediums, do AI queries ultimately converge on a centralized, highly consolidated set of CDN edge ASNs (e.g., Cloudflare)?
- **RQ3 (Jurisdictional Traversal):** How does jurisdictional traversal vary for AI queries originating from regions selected for their diverse economic and regulatory profiles (like India, Pakistan, Kenya, and France), and to what extent are these queries routed through foreign jurisdictions?

2 Hypothesis

We hypothesize that the physical infrastructure supporting generative AI is hyper-centralized, yet highly dependent on the origin network and region. For path divergence (RQ1), we hypothesize that the choice of local ISP fundamentally alters the initial transit architecture; specifically, different transport mediums (e.g., cellular versus residential broadband) will expose AI queries to varying hop counts, distinct intermediate data handlers, and fundamental protocol shifts, such as a heavy reliance on internal IPv6 routing by cellular carriers versus mixed IPv4/IPv6 transit by traditional ISPs. Despite this initial structural divergence, we hypothesize high infrastructural convergence (RQ2), where all North American queries rapidly collapse onto a narrow, consolidated set of domestic edge ASNs. Finally, for global jurisdictional traversal (RQ3), we hypothesize that queries originating from regions with differing economic profiles and network regulations will frequently experience anomalous routing—bypassing local peering exchanges and crossing into foreign jurisdictions governed by stringent network and surveillance policies (e.g., the UK or US)—highlighting the uneven global distribution of localized AI edge nodes.

3 Related Work

Current literature heavily examines the algorithmic capabilities of Large Language Models (LLMs), but there is a notable gap regarding their physical network reliance. Previous studies on cloud consolidation [1] establish that modern web traffic is heavily centralized. Similarly, work utilizing CAIDA BGP datasets has documented routing inefficiencies and BGP tromboning in standard web traffic.

Furthermore, mapping infrastructural dependencies requires translating IP-level paths into accurate Autonomous System (AS) and organizational topologies. Previous literature highlights that deriving precise IP-to-AS mappings is an ongoing challenge, as there is currently no completely accurate or universally accepted mapping methodology [7]. While BGP tables are frequently used to infer network topologies, the AS paths advertised via BGP reflect complex, localized administrative policies and often differ from the actual forwarding paths that data packets traverse [6]. To mitigate these mapping inaccuracies, researchers often pair active traceroute measurements (the primary tool for discovering end-to-end paths) with BGP data collected from multiple vantage points [6, 7].

4 Theoretical contribution

Our primary theoretical contribution is providing empirical, quantitative data to evaluate both domestic routing disparities and global geopolitical data flow imbalances within the generative AI ecosystem.

Domestically, by addressing path divergence, we aim to document how consumer ISP choice influences the intermediate network topology of an AI query. We map the extent to which disparate access mediums expose users to varying hop counts, differing internal ISP traffic engineering protocols and distinct intermediate data handlers. Furthermore, by addressing infrastructural convergence, we investigate whether these divergent initial paths ultimately collapse onto a narrow set of CDN chokepoints. This approach highlights a complex North American topology: we explore if ISP choice significantly alters the transit journey and intermediary exposure, and whether the destination remains highly consolidated, which would pose centralized domestic data-collection risks.

Additionally, by globally observing how varying economic profiles and stringent national network policies correlate with the physical international routing of AI data, this work provides foundational empirical evidence for current regulatory debates regarding data sovereignty, international privacy disparities, and mass surveillance vulnerabilities inherent in cloud-based AI.

5 Method

We utilize active network measurements to map both a controlled American baseline and a diverse global dataset of AI traffic routing.

Tech Stack:

- **Collection:** Python 3 ('subprocess', 'requests') automating Windows 'tracert -d' for local measurements.
- **Global Expansion:** RIPE Atlas REST API for orchestrated ICMP traceroutes from global vantage points.
- **Enrichment:** We will use the CAIDA AS2Org dataset to map raw IP addresses to their respective corporate organizations. Alternatively, if dynamic resolution is required, we will utilize BGP data from RouteViews or RIS to enrich the traceroute paths with their parent ASNs.

Execution:

- **W**e will execute automated, continuous traceroutes to target AI endpoints using distinct American access mediums (residential broadband vs. cellular). This phase captures the initial AS paths to measure variances in hop counts and internal ISP routing behavior (e.g., IPv4 vs. IPv6 transit).
- **U**sing the same American baseline dataset, we will trace the queries to their final destination ASNs. This execution phase maps the terminal endpoints to observe whether the divergent initial paths structurally collapse onto an identical, centralized set of CDN edge nodes.
- **W**e will programmatically dispatch one-off RIPE Atlas probes from specific ASNs in India, Pakistan, South Africa, and the UK. We will then cross-reference the resulting IP hops with geographic databases to physically trace the movement of packets across international borders, documenting which foreign regulatory jurisdictions the data enters prior to reaching the AI service.

6 Analysis and/or systems building plan

The resulting JSON trace datasets will be parsed to construct directed AS-level graphs. We will analyze the frequency of specific transit ASNs to identify chokepoints and measure the exact delta in AS path length and geographic distance between the US baseline and the RIPE Atlas probes.

- **Member 1, Member 2:** Manages the American baseline data collection (RQ1/RQ2), maintaining the Python ISP rotation scripts and executing the local Windows `tracert` cycles across different access mediums. Orchestrates the RIPE Atlas REST API integration, handling the programmatic selection and deployment of probes across India, Pakistan, South Africa, and the UK (RQ3).
- **Member 3, Member 4:** Develops the BGP enrichment pipeline, responsible for parsing the raw traceroute JSONs and integrating the CAIDA AS2Org and RouteViews datasets to map IPs to their parent organizations. Synthesizes the enriched datasets to generate directed AS-level topology graphs, performs the cross-jurisdictional traversal analysis, and measures the path length deltas between the US and global baselines.

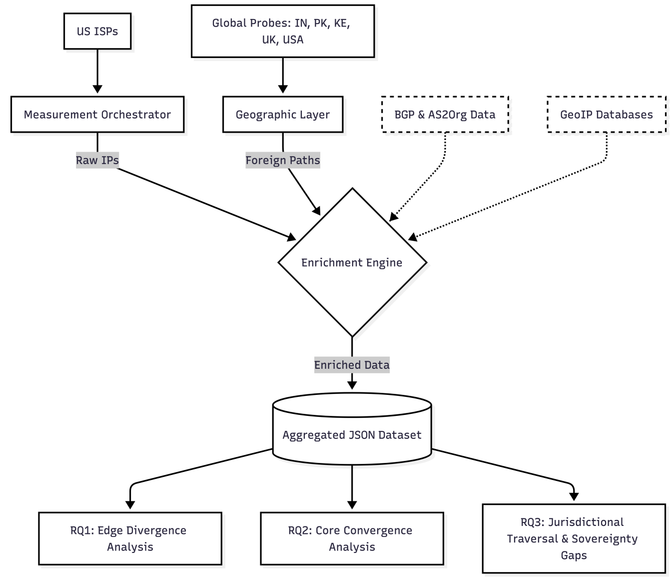

Figure 1: System diagram

Expected Milestones:

- **Week 7:** Finalize local data collection scripts; begin US baseline measurements alongside Deploy RIPE Atlas probes to target international regions; download CAIDA/RouteViews datasets.
- **Week 8:** Build Python parsers to map raw IP traces to AS2Org BGP data.
- **Week 9:** Generate topology graphs; perform cross-jurisdictional traversal analysis to compare US convergence against global jurisdictional routing and Synthesize results and finalize the project paper.

7 Biggest risk

The Risk: Incomplete Path Data via ICMP Rate Limiting. When mapping individual IP hops to their parent ASes, we will encounter gaps. Many massive transit routers are configured to drop standard traceroute probes, resulting in hidden hops ($\star\ \star\ \star$) that obscure the true transit path and obscure potential jurisdictional crossings.While we utilize the CAIDA AS2Org dataset and BGP data from RouteViews or RIS to enrich our IP mappings and infer hidden hops, these external datasets carry inherent risks.

The Mitigation: We will actively preserve these null hops in our JSON data structures to maintain mathematically accurate path lengths. To resolve these dataset limitations and ensure high-fidelity mapping, we will employ a multi-layered cross-validation strategy.

References

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